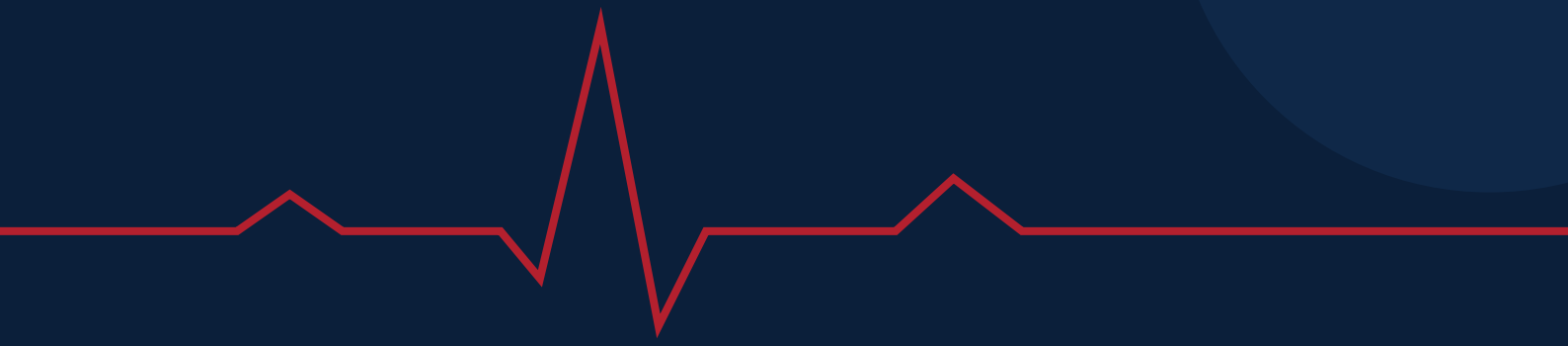
ADULT ECHOCARDIOGRAPHY | FELLOW & CONSULTANT REFERENCE

ECHO ESSENTIALS

The complete guideline-based companion



Chamber quantification and LV function | Diastology and HFpEF
Right heart and pulmonary hypertension
Aortic stenosis: full grading, critical AS, both low-flow types
AR, MS, MR, TR | Reporting checklist | Teaching pearls

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Aligned with ASE / EACVI recommendations and the 2025 ESC/EACTS valvular heart disease guidelines.

WELCOME

How to use this book

Echo Essentials is built for the busy fellow and the practising echocardiographer. Every chapter follows the same rhythm: **the number**, **the reason**, **the trap**. You get the thresholds you will quote at the reading station, the logic behind them, and the errors that fool even experienced readers.

SOURCES AND VERSION CONTROL

Thresholds follow the **ASE/EACVI** recommendations (chamber quantification 2015; native valve regurgitation 2017; native valve stenosis 2017; ASE 2025 diastolic function and HFpEF update; ASE/EACVI strain consensus) and the **2025 ESC/EACTS valvular heart disease guidelines**, with ACC/AHA 2020 shown where the two societies differ.

Where societies disagree, both values are shown and labelled. Intervention classes change quickly: confirm against the full guideline text before using a class of recommendation in a report or an exam.

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READING KEY

Red = high-stakes threshold or common error. **Gold** = version and source notes. **Green** = clinical pearl for teaching.

Abbreviations: Vmax peak aortic jet velocity; MG mean gradient; AVA aortic valve area; AVAi indexed AVA; DVI dimensionless velocity index; SVi stroke volume index; LFLG low-flow low-gradient; DSE dobutamine stress echo; EROA effective regurgitant orifice area; RVol regurgitant volume; RF regurgitant fraction.

CHAPTER 1

Chamber quantification and LV systolic function

Measure it the same way every time and the numbers become comparable.

The foundation

- Use **biplane Simpson's method** for LV volumes and LVEF; add contrast when two or more contiguous segments are not seen. Avoid foreshortening.
- Use **linear dimensions** from the parasternal long axis (2D-guided), at end-diastole (EDD) and end-systole (ESD), perpendicular to the LV long axis.
- Report **indexed** values (BSA) for volumes, mass and left atrial size.

Reference limits (ASE/EACVI 2015)

Parameter	Men: normal	Women: normal	Comment
LVEF (biplane)	52 - 72%	54 - 74%	Mild 41-51 / 41-53; moderate 30-40; severe < 30
LVEDD	42 - 58 mm	38 - 52 mm	Linear, PLAX
LVESD	25 - 40 mm	22 - 35 mm	
LV mass index (2D)	50 - 102 g/m²	44 - 88 g/m²	Linear method: 49-115 / 43-95
Relative wall thickness	≤ 0.42	≤ 0.42	> 0.42 = concentric
LA volume index	16 - 34 mL/m²	16 - 34 mL/m²	Mild 35-41, mod 42-48, severe > 48
LV GLS (vendor dependent)	about -18 to -22%	about -19 to -23%	Abnormal when less negative than about -16%

LV geometry in one line

- **Normal geometry**: mass normal, RWT ≤ 0.42. **Concentric remodelling**: mass normal, RWT > 0.42. **Concentric hypertrophy**: mass raised, RWT > 0.42. **Eccentric hypertrophy**: mass raised, RWT ≤ 0.42.

TEACHING PEARL: WHY GLS EARNS ITS PLACE

GLS detects subclinical dysfunction before LVEF falls: cardio-oncology (a relative fall of > 15% from baseline is clinically significant), amyloid (apical sparing), and aortic stenosis with preserved EF. Use one vendor, one reader, the same frame rate, and report the vendor.

COMMON ERROR

Reporting LVEF from a **foreshortened** apical view or from a single plane. Eyeball EF is acceptable only with experience and should be cross-checked against Simpson's or contrast.

CHAPTER 2

Diastolic function and HFpEF

Estimate left atrial pressure. Grade second.

ASE 2025 update: what changed

- The July 2025 ASE document restructures the approach: **first decide whether left atrial pressure (LAP) is normal or raised**, then grade dysfunction, and it adds explicit guidance on **diagnosing HFpEF**.
- Always interpret in context: LVEF, atrial rhythm, mitral valve disease, annular calcification, prosthetic valves, LBBB and pacing all change how parameters behave.
- **Verify the stepwise algorithm against the published ASE 2025 flow chart**; the core parameters below are the building blocks it uses.

Core parameters and abnormal cut-offs

Parameter	Abnormal / suggests raised LAP	Notes
Septal e'	< 7 cm/s	Age-dependent; lower with MAC
Lateral e'	< 10 cm/s	
Average E/e'	> 14	Septal > 15, lateral > 13
TR peak velocity	> 2.8 m/s	Needs a good CW signal
LA volume index	> 34 mL/m²	Chronic exposure marker; raised in AF and athletes
E/A and E velocity	E/A ≥ 2 suggests raised LAP; E/A ≤ 0.8 with E ≤ 50 cm/s suggests normal LAP	Intermediate E/A needs the other parameters
Pulmonary vein S/D, Ar - A duration	S/D < 1; Ar - A > 30 ms	Supportive

Grading in practice

- **Grade I**: impaired relaxation, normal LAP. **Grade II**: pseudonormal, raised LAP. **Grade III**: restrictive filling, markedly raised LAP.
- **Atrial fibrillation**: use septal E/e' ≥ 11, deceleration time ≤ 160 ms, IVRT ≤ 65 ms, and TR velocity > 2.8 m/s.
- **Valsalva**: a fall in E/A of ≥ 0.5 unmasks pseudonormal filling.

TEACHING PEARL

Ask the fellow: *"Is the left atrial pressure raised, and how do you know?"* The grade is a label. The pressure is the answer that changes management.

COMMON ERROR

Using E/e' in moderate-to-severe MAC, mitral stenosis, prosthetic mitral valve or severe primary MR. In these settings use other markers (IVRT, pulmonary vein flow, TR velocity, strain).

CHAPTER 3

The right heart and pulmonary hypertension

The right ventricle is an unforgiving window into the patient.

Parameter	Abnormal if	Comment
TAPSE	< 17 mm	Longitudinal function; angle dependent
RV S' (tissue Doppler)	< 9.5 cm/s	Basal free wall
RV FAC	< 35%	Needs full endocardial tracing
RV free wall longitudinal strain	Less negative than -20%	Excludes the septum
RV basal diameter	> 41 mm	RV-focused apical view
RA area	> 18 cm²	End-systole
RIMP (TDI)	> 0.54	Unreliable when RA pressure is high

Estimating pulmonary pressure

PASP = 4 × (TR Vmax)² + estimated RAP. Estimate RAP from the IVC: **3 mmHg** if ≤ 21 mm and collapses > 50%; **15 mmHg** if > 21 mm and collapses < 50%; **8 mmHg** for the intermediate pattern.

Echocardiographic probability of PH (ESC/ERS 2022)

TR Vmax	Other PH signs	Probability
≤ 2.8 m/s or not measurable	None	Low
≤ 2.8 m/s	Present	Intermediate
2.9 - 3.4 m/s	Absent	Intermediate
2.9 - 3.4 m/s	Present	High
> 3.4 m/s	Not required	High

- **Other PH signs**: RV/LV basal ratio > 1.0, flattened septum (LV eccentricity index > 1.1), RVOT acceleration time < 105 ms or mid-systolic notching, early diastolic PR velocity > 2.2 m/s, PA diameter > 25 mm, IVC > 21 mm with reduced collapse, RA area > 18 cm².
- TAPSE / PASP ratio < 0.55 mm/mmHg identifies RV-pulmonary arterial uncoupling and carries a worse prognosis.

> **COMMON ERROR**
> Calling PASP from a poorly Dopplered, **underestimated** TR envelope. A faint CW signal gives a falsely low velocity. Enhance with agitated saline if needed, and say so in the report.

CHAPTER 4

Aortic stenosis: the complete chapter

Grade it, define it, and never miss the low-flow patient.

Step 1: acquire the three core measurements

- **Vmax**: CW Doppler from multiple windows (apical, right parasternal, suprasternal) with a dedicated non-imaging probe. Report the **highest** signal. Angle and alignment errors always *underestimate*.
- **Mean gradient**: planimetered from the same CW envelope, using the Bernoulli equation ($4V^2$).
- **AVA by continuity**: AVA = $(\pi \times (\text{LVOT diameter} / 2)^2 \times \text{VTI}_{\text{LVOT}}) / \text{VTI}_{\text{AV}}$. The LVOT diameter is squared, so a 1 mm error shifts AVA by about 0.2 cm².
- **DVI** = $\text{VTI}_{\text{LVOT}} / \text{VTI}_{\text{AV}}$. It avoids the LVOT diameter. Severe if ≤ 0.25.
- **Stroke volume index** = (LVOT area × VTI_{LVOT}) / BSA. Low flow = **SVi < 35 mL/m²**.

Step 2: grade severity

	Mild	Moderate	Severe	Very severe / critical
Vmax (m/s)	2.6 - 2.9	3.0 - 3.9	≥ 4.0	≥ 5.0
Mean gradient (mmHg)	< 20	20 - 39	≥ 40	≥ 60
AVA (cm²)	> 1.5	1.0 - 1.5	≤ 1.0	typically < 0.6
AVAi (cm²/m²)	> 0.85	0.60 - 0.85	≤ 0.6	
DVI	> 0.50	0.25 - 0.50	≤ 0.25	

CRITICAL AORTIC STENOSIS: HOW THE TERM IS USED

Adult practice: **critical** (also called **very severe**) AS = **Vmax ≥ 5.0 m/s and/or mean gradient ≥ 60 mmHg**, usually with AVA < 0.6 cm². ACC/AHA 2020 uses "very severe"; ESC/EACTS lists Vmax > 5.0 m/s as a high-risk marker that supports early intervention even without symptoms.

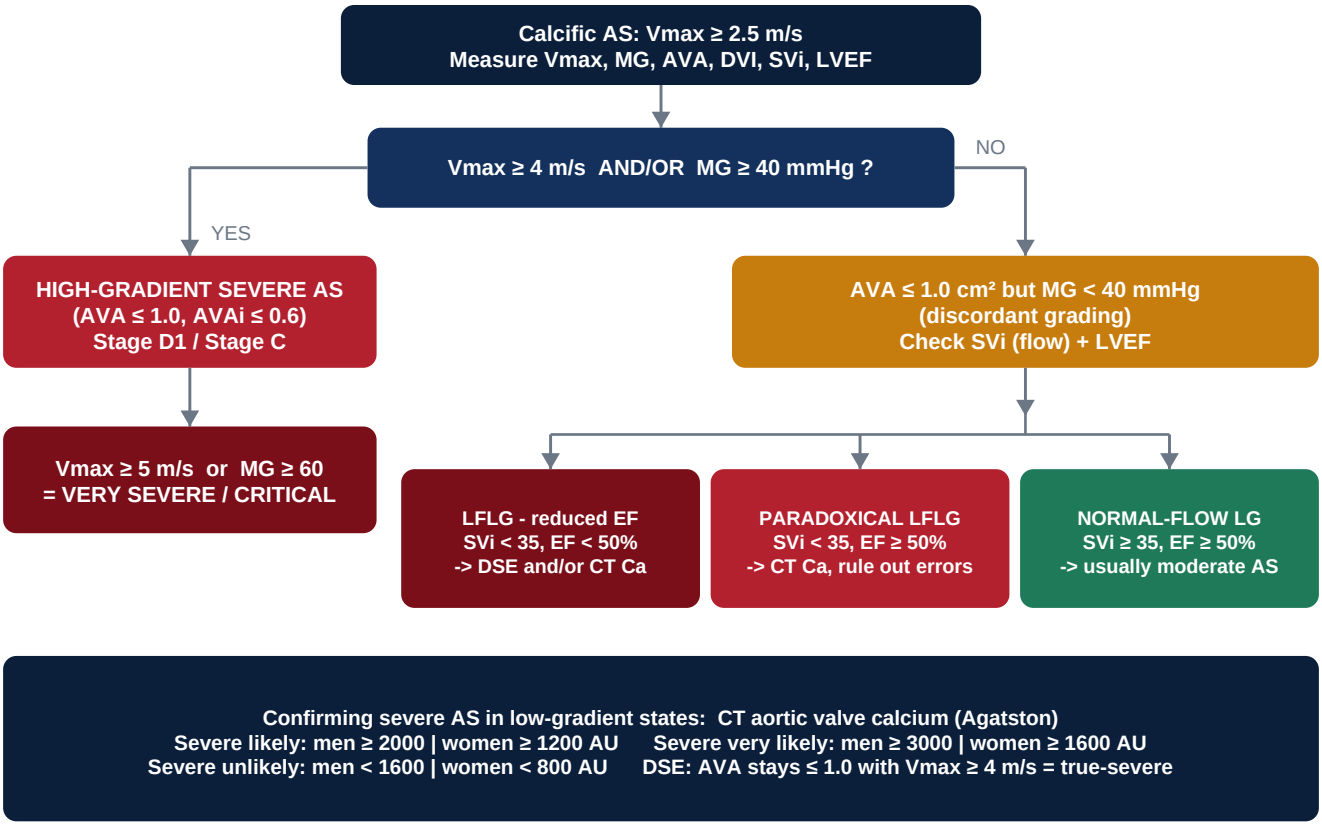
No society defines "critical" by a single universal number. State the numbers you used in the report instead of the label alone.

Neonatal critical AS is a different entity: a duct-dependent systemic circulation presenting in the first days of life. Do not apply adult cut-offs.

Remember that a very high gradient needs adequate flow and a normal LVEF. A patient can be critically stenosed with a "modest" gradient when stroke volume is low.

Step 3: resolve discordance

About one third of patients with AVA ≤ 1.0 cm² have a mean gradient below 40 mmHg. Start from the flow: **SVi**, then **LVEF**.



Confirming severe AS in low-gradient states: CT aortic valve calcium (Agatston)
Severe likely: men ≥ 2000 | women ≥ 1200 AU **Severe very likely: men ≥ 3000 | women ≥ 1600 AU**
Severe unlikely: men < 1600 | women < 800 AU **DSE: AVA stays ≤ 1.0 with Vmax ≥ 4 m/s = true-severe**

The two types of low-flow, low-gradient AS

	Classical LFLG (reduced EF)	Paradoxical LFLG (preserved EF)
Definition	AVA ≤ 1.0 cm², MG < 40 mmHg, **LVEF < 50%**, SVi < 35 mL/m²	AVA ≤ 1.0 cm² (AVAi ≤ 0.6), MG < 40 mmHg, **LVEF ≥ 50%**, SVi < 35 mL/m²
ACC/AHA stage	D2	D3
Typical patient	Dilated LV, ischaemic or dilated cardiomyopathy; AS may be true or pseudo-severe	Older, often women; small hypertrophied LV, hypertension, restrictive physiology, impaired GLS, atrial fibrillation
Key question	Is the valve truly severe, or is a weak ventricle unable to open a moderately stenotic valve?	Is the low flow real, or is it a measurement problem (LVOT area, BSA, blood pressure)?
First test	Low-dose **dobutamine stress echo** and/or **CT calcium score**	Exclude errors first, then **CT calcium score**; DSE has no role to raise flow
Outcome if untreated	Poor; absence of flow reserve is worse still	Poor; mortality similar to high-gradient severe AS
Management	AVR recommended when severe AS is confirmed (Class I in guidelines); TAVI preferred if high surgical risk	AVR considered in symptomatic patients once alternative explanations are excluded

Dobutamine stress echo in classical LFLG

- Infuse dobutamine from 5 up to a maximum of **20 µg/kg/min**. Measure SV, Vmax, MG and AVA at each stage.
- **Flow (contractile) reserve** = stroke volume rise of ≥ 20%.
- **True-severe AS**: AVA remains ≤ 1.0 cm² with Vmax ≥ 4 m/s (or MG ≥ 40 mmHg) at the higher flow.

- **Pseudo-severe AS**: AVA rises above 1.0-1.2 cm² as flow normalises and the gradient stays low.
- **No flow reserve**: the test is non-diagnostic. Operative risk is higher, but benefit from AVR persists. Use CT calcium scoring to adjudicate.

CT aortic valve calcium score (non-contrast)

	Men	Women
Severe AS very likely	≥ 3000 AU	≥ 1600 AU
Severe AS likely	≥ 2000 AU	≥ 1200 AU
Severe AS unlikely	< 1600 AU	< 800 AU

Use the sex-specific thresholds in the ESC/EACTS guideline. Calcium score confirms a calcific burden; it does not replace haemodynamics, and severe AS can occur with less calcium, particularly in younger or rheumatic or bicuspid valves.

Paradoxical LFLG: the checklist before you believe it

- **Blood pressure**: scan when systolic BP is controlled (< 140 mmHg). Hypertension reduces flow and distorts gradients.
- **LVOT area**: the LVOT is elliptical. 2D diameter underestimates area. Cross-check with 3D echo or CT planimetry. Underestimated area gives falsely low AVA and SV.
- **Body size**: use AVAi in small women. An AVA of 0.9 cm² in a very small patient may be moderate.
- **Heart rhythm**: average beats in AF (5 consecutive cycles).
- **Associated disease**: significant MR, AR, MS or amyloid can mimic the picture. Pressure recovery can falsely lower gradients in small aortas (< 30 mm).

Normal-flow, low-gradient AS

AVA ≤ 1.0 cm² with MG < 40 mmHg, SVi ≥ 35 mL/m² and LVEF ≥ 50%. This is usually **moderate** AS or a measurement inconsistency (the AVA cut-off and the gradient cut-off do not correspond exactly). Reassess measurements, then follow clinically.

Step 4: from numbers to decisions

Intervention indications below summarise the common ground between ESC/EACTS and ACC/AHA. **Always check the Class and Level in the current full text**: early-intervention evidence (EARLY TAVR, AVATAR, RECOVERY) is shifting practice.

Situation	Typical recommendation
Symptomatic severe high-gradient AS	Intervention (Class I). Choice of TAVI vs SAVR by age, life expectancy, anatomy, surgical risk and Heart Team
Symptomatic severe LFLG, reduced EF	Intervention (Class I) once severity is confirmed (DSE or CT calcium)
Symptomatic severe LFLG, preserved EF	Intervention if severity confirmed and no alternative explanation for symptoms
Asymptomatic, LVEF < 50% without another cause	Intervention (Class I in 2021 ESC; check 2025 text for the LVEF cut-off)
Asymptomatic, positive exercise test (symptoms or fall in BP)	Intervention (Class I in guidelines; test is contraindicated if symptomatic)
Asymptomatic very severe (Vmax ≥ 5 m/s, MG ≥ 60)	Intervention considered in low-risk patients (Class IIa)
Asymptomatic with BNP > 3x normal, or rapid progression (Vmax rise > 0.3 m/s/year), or severe calcification	Intervention considered (Class IIa) - see current guideline for exact wording
Mild or moderate AS	Surveillance; moderate AS with LV dysfunction is an emerging research area

SURVEILLANCE INTERVALS (TYPICAL)

Mild AS: every 3-5 years. Moderate: every 1-2 years. Severe (asymptomatic): every 6-12 months and educate the patient about symptoms. Shorten the interval if Vmax > 4.5 m/s, calcification is heavy, or progression is fast.

TEACHING PEARL: THE FELLOW'S FIVE-LINE AS REPORT

1. Valve morphology (tricuspid / bicuspid / rheumatic) and calcification. 2. Vmax, MG, AVA, AVAi, DVI. 3. LVOT diameter and SVi (flow). 4. LVEF and GLS, LV mass, RWT. 5. A single sentence on severity, flow state, and what would change management.

CHAPTER 5

Aortic regurgitation

Integrate. No single parameter is enough.

Parameter	Mild	Moderate	Severe
Vena contracta (mm)	< 3	3 - 6	> 6
Jet width / LVOT (central jet)	< 25%	25 - 64%	≥ 65%
Pressure half-time (ms)	> 500	200 - 500	< 200
Descending aorta flow reversal	Brief, early diastole	Intermediate	Holodiastolic (end-diastolic velocity > 20 cm/s in the abdominal aorta)
EROA (cm²)	< 0.10	0.10 - 0.29	≥ 0.30
Regurgitant volume (mL)	< 30	30 - 59	≥ 60
Regurgitant fraction (%)	< 30	30 - 49	≥ 50
LV size	Normal	Normal or dilated	Usually dilated in chronic AR

Timing of surgery (chronic severe AR)

- Symptomatic: surgery (Class I).
- Asymptomatic with **LVEF ≤ 50% (ESC 2021) / ≤ 55% (ACC/AHA 2020)**: surgery. Check the 2025 ESC wording.
- Asymptomatic with **LVESD > 50 mm** (or > 25 mm/m² BSA in small patients, ESC) or LVEDD > 70 mm: surgery should be considered.
- Always measure the **aortic root and ascending aorta**: if the root is dilated (>= 45 mm in Marfan, >= 50 mm in bicuspid disease with risk factors), the threshold for surgery changes.

PEARL

Acute severe AR: the LV is small, the PHT is short (the pressure equalises quickly), and the diastolic MR with early mitral valve closure is a clue. These patients need urgent surgery, not echo trending.

COMMON ERROR

Underestimating eccentric jets with colour jet area. Use **vena contracta** and PISA / flow reversal instead.

CHAPTER 6

Mitral stenosis

Rheumatic mitral stenosis is not gone. Look for it.

Parameter	Mild	Moderate	Severe (clinically significant)
MVA (planimetry, cm²)	> 1.5	1.5 - 1.0 (ESC: ≤ 1.5 = significant)	≤ 1.5; very severe ≤ 1.0
Pressure half-time (ms)	< 150	150 - 219	≥ 150; very severe ≥ 220
Mean gradient	< 5	5 - 10	> 10 at HR 60-80 (very heart-rate dependent)
PASP (mmHg)	< 30	30 - 50	> 50

Technique

- **MVA by 2D planimetry** in the parasternal short axis at the leaflet tips, lowest gain that shows the orifice. The gold standard when feasible.
- **PHT**: MVA = 220 / PHT. Inaccurate with AF, tachycardia, abnormal LA / LV compliance, after balloon valvotomy, and with significant AR.
- **Continuity or PISA** if windows are poor. Record the heart rate with the gradient.
- **Wilkins score**: mobility, thickening, calcification and subvalvular thickening, each 1-4. A score of **≤ 8** favours percutaneous balloon commissurotomy; commissural calcification and severe subvalvular disease are adverse.

Before percutaneous commissurotomy

- Exclude **left atrial / LAA thrombus** (TEE) and significant MR (more than mild).
- Intervention is indicated in symptomatic patients with clinically significant MS (MVA ≤ 1.5 cm²) and favourable anatomy; consider in asymptomatic patients with high thromboembolic risk or pulmonary hypertension (PASP > 50 mmHg).

PEARL: DEGENERATIVE MS
Mitral annular calcification extending into the leaflets can cause stenosis with a high gradient but a small area by PHT being unreliable. Use planimetry (often impossible) or the continuity method and think of the CT.

CHAPTER 7

Mitral regurgitation

First ask: primary or secondary? The answer changes the thresholds.

Parameter	Mild	Moderate	Severe
Vena contracta (mm)	< 3	3 - 6.9	≥ 7 (3D VCA ≥ 0.4 cm²)
EROA, primary (cm²)	< 0.20	0.20 - 0.39	≥ 0.40
Regurgitant volume (mL)	< 30	30 - 59	≥ 60
Regurgitant fraction (%)	< 30	30 - 49	≥ 50
PISA radius (Nyquist 40 cm/s)	< 4 mm	4 - 9 mm	≥ 10 mm
Pulmonary vein flow	Systolic dominance	Blunted	Systolic flow reversal
Mitral E velocity	-	-	> 1.5 m/s; MV VTI / LVOT VTI > 1.4

Secondary (functional) MR

- Severe if **EROA ≥ 0.20 cm² or RVol ≥ 30 mL** in ESC/ACC practice (the orifice is elliptical and the thresholds are lower), and always correlate with symptoms and LV remodelling. COAPT enrolled patients with EROA > 0.30 cm² (or RVol > 45 mL).
- Transcatheter edge-to-edge repair is considered in symptomatic patients on optimal guideline-directed therapy with suitable anatomy; follow the COAPT-based selection criteria (LVEF 20-50%, LVESD ≤ 70 mm, PASP ≤ 70 mmHg).

Primary MR: when to operate

- Symptomatic severe MR with LVEF > 30%: surgery (repair preferred when durable).
- Asymptomatic with **LVESD ≥ 40 mm** or **LVEF ≤ 60%**: surgery.
- Asymptomatic with **new AF**, **PASP > 50 mmHg** at rest, or LA volume index ≥ 60 mL/m²: surgery should be considered (repair, in a heart valve centre with high repair rate).

COMMON ERROR

Underestimating **eccentric wall-hugging jets** by colour jet area (Coanda effect). Use vena contracta, pulmonary vein flow, PISA and quantitative volumes, and move to TEE or CMR when in doubt.

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CHAPTER 8

Tricuspid regurgitation

Severe is no longer the top of the scale.

Parameter	Mild	Moderate	Severe	Massive	Torrential
Vena contracta (mm)	< 3	3 - 6.9	7 - 13	14 - 20	≥ 21
EROA (mm²)	< 20	20 - 39	40 - 59	60 - 79	≥ 80
Regurgitant volume (mL)	< 30	30 - 44	45 - 59	60 - 74	≥ 75
PISA radius (mm, Nyquist 28 cm/s)	≤ 5	6 - 9	> 9		

- Supportive severe signs: dense triangular CW signal with early peaking, hepatic vein systolic flow reversal, E velocity > 1.0 m/s, RA / RV / IVC dilatation, annular dilation > 40 mm (or > 21 mm/m²).
- The expanded **massive and torrential** grades (Hahn et al.) matter because they stratify outcome and select patients for transcatheter tricuspid therapy.
- Classify the mechanism: **secondary** (annular dilation, most common), **primary**, or **lead-related**. Report RV size and function and the rhythm.

PEARL

A "free-flowing" TR has a low velocity and an early-peaking, triangular CW signal. A low TR velocity does **not** mean a low pulmonary pressure; it means equalised pressures.

CHAPTER 9

The reporting checklist

A report is a message to a decision-maker.

A structure consistent with the ASE 2025 adult echo reporting standards

- **Indication and clinical question**, image quality, rhythm and heart rate, BP and BSA.
- **LV**: size, wall thickness / mass, geometry, EF method, regional wall motion, GLS if done.
- **Diastology**: is LAP raised? Grade and parameters used.
- **RV**: size, function (TAPSE, S', FAC, FWLS), PASP and probability of PH.
- **Atria**, IVC, pericardium, aorta (root, ascending), venous structures.
- **Valves**: for each lesion give mechanism, severity with the parameters used, and the consequence for the chambers.
- **Conclusion**: three to five lines, the first line answers the clinical question. Compare with prior studies.

CHECKLIST BEFORE YOU SIGN

1. Did I answer the referral question? 2. Are LVEF and volumes indexed and compared with the last study? 3. If AS: SVi reported, flow state named? 4. If MR / TR: mechanism and quantitative parameters listed? 5. Is RAP, PASP and PH probability reported? 6. Are all measurements in the units and with the cut-offs of the current guidelines?

CHAPTER 10

Teaching pearls and references

Take these to the next case conference.

- **Always say the flow.** In any discordant AS case, the SVi is the first number you report after the gradient.
- **Gradient, area, flow, EF.** Four numbers, one story. If the story does not fit, a measurement is wrong.
- **Use the right probe.** A dedicated Pedoff probe and a systematic search can change a moderate AS into a severe one.
- **Do not trust a single jet.** Regurgitation needs mechanism, qualitative, semi-quantitative and quantitative parameters.
- **Think of the mimic.** Amyloid, hypertrophic cardiomyopathy and hypertension can all present as low-flow, low-gradient AS or pseudo-diastology.
- **Correlate with the patient.** Echo thresholds are guides; symptoms and the Heart Team decide.

Key references

- Lang RM et al. Recommendations for Cardiac Chamber Quantification by Echocardiography in Adults. J Am Soc Echocardiogr 2015;28:1-39.
- Baumgartner H et al. Recommendations on the Echocardiographic Assessment of Aortic Valve Stenosis (EACVI/ASE). J Am Soc Echocardiogr 2017;30:372-392.
- Zoghbi WA et al. Recommendations for Noninvasive Evaluation of Native Valvular Regurgitation. J Am Soc Echocardiogr 2017;30:303-371.
- ASE. Recommendations for the Evaluation of LV Diastolic Function by Echocardiography and for HFpEF Diagnosis. J Am Soc Echocardiogr 2025 (July issue).
- Vahanian A et al. 2025 ESC/EACTS Guidelines for the management of valvular heart disease.
- Otto CM et al. 2020 ACC/AHA Guideline for the Management of Patients With Valvular Heart Disease.
- Hahn RT, Zamorano JL. The need for a new tricuspid regurgitation grading scheme. Eur Heart J Cardiovasc Imaging 2017;18:1342-1343.
- Humbert M et al. 2022 ESC/ERS Guidelines for pulmonary hypertension.
- ASE. Guidelines for the Standardization of Adult Echocardiography Reporting. J Am Soc Echocardiogr 2025;38:735-774.
- Clavel MA, Pibarot P et al. Sex-specific aortic valve calcium thresholds in aortic stenosis (JACC 2013 and later work); Blais C et al. Projected valve area at normal flow rate. Circulation 2006.

DISCLAIMER

For educational use by qualified clinicians. Not a substitute for the full published guidelines, local protocols, or clinical judgement. Always verify thresholds against the current ASE, EACVI and ESC/EACTS documents.

This is an educational summary. Numbers have been checked against the guideline documents listed, but the reader is responsible for confirming current versions before clinical or examination use.